



DW01_prienc

Priority Encoder

Version, STAR, and myDesignWare Subscriptions: [IP Directory](#)

Features and Benefits

- Parameterized word length
- Inferable using a function call

Revision History



Description

DW01_prienc encodes the input port `A` to a binary value on output port `INDEX`. The encoded value of `A` is determined by the bit position of the most significant 1 bit. All bits on `A` lower than the most significant 1 bit are “don’t care”.

The acceptable values for `INDEX_width` are greater than or equal to $\text{ceil}(\log_2[A_width])$, as described in [Table 1-2](#). However, the recommended value for `INDEX_width` is $\text{ceil}(\log_2[A_width + 1])$. With this value, the output `INDEX` can cover all the possible legal values. Note that if `INDEX_width` is equal to $\text{ceil}(\log_2[A_width])$ with `A_width` greater than 1, all 0s on pin `A` would result in the same `INDEX` output value as when a 1 is on the most significant bit of pin `A`.

Table 1-1 Pin Description

Pin Name	Width	Direction	Function
A	<i>A_width</i>	Input	Input data
INDEX	<i>INDEX_width</i>	Output	Binary encoded output data

Table 1-2 Parameter Description

Parameter	Values	Description
A_width	≥ 1	Word length of input <code>A</code>
INDEX_width	$\geq \text{ceil}(\log_2[A_width])$	Word length of output <code>INDEX</code>

Table 1-3 Synthesis Implementations

Implementation Name	Function	License Feature Required
str	Synthesis model	DesignWare
cla ^a	Synthesis model	DesignWare
aot	Synthesis model	DesignWare

- a. The 'cla' implementation is only available when $A_width \leq 512$.

Table 1-4 Simulation Models

Model	Function
DW01.DW01_PRIENC_CFG_SIM	Design unit name for VHDL simulation
dw/dw01/src/DW01_prienc_sim.vhd	VHDL simulation model source code
dw/sim_ver/DW01_prienc.v	Verilog simulation model source code

Table 1-5 Truth Table ($A_width = 8$, $INDEX_width = 4$)

A(7)	A(6)	A(5)	A(4)	A(3)	A(2)	A(1)	A(0)	INDEX(3:0)
0	0	0	0	0	0	0	0	0000
0	0	0	0	0	0	0	1	0001
0	0	0	0	0	0	1	X	0010
0	0	0	0	0	1	X	X	0011
0	0	0	0	1	X	X	X	0100
0	0	0	1	X	X	X	X	0101
0	0	1	X	X	X	X	X	0110
0	1	X	X	X	X	X	X	0111
1	X	X	X	X	X	X	X	1000

Related Topics

- [Logic – Combinational Overview](#)
- [DesignWare Building Blocks User Guide](#)

HDL Usage Through Function Inferencing - VHDL

```
library IEEE, DWARE;
use IEEE.std_logic_1164.all;
use IEEE.std_logic_arith.all;
use DWARE.DW_foundation_arith.all;

entity DW01_prienc_func is
  generic(func_A_width : integer := 8; func_INDEX_width : integer := 4);
  port(func_A: in std_logic_vector(func_A_width-1 downto 0);
        INDEX_func, INDEX_func_TC, INDEX_func_UNUS: out
          std_logic_vector(func_INDEX_width-1 downto 0));
end DW01_prienc_func;

architecture func of DW01_prienc_func is
begin

  INDEX_func <= DWF_prienc(func_A, func_INDEX_width);
  INDEX_func_TC <= std_logic_vector(DWF_prienc(signed(func_A),
                                              func_INDEX_width));
  INDEX_func_UNUS <= std_logic_vector(DWF_prienc(unsigned(func_A),
                                              func_INDEX_width));

end func;
```

HDL Usage Through Function Inferencing - Verilog

```
module DW01_prienc_func (func_A, INDEX_func);  
    parameter func_A_width = 8;  
    parameter func_INDEX_width = 4;  
  
    // Passes the widths to the prienc function  
    parameter A_width = func_A_width;  
    parameter INDEX_width = func_INDEX_width;  
  
    // Please add search_path = search_path + {synopsys_root + "/dw/sim_ver"}  
    // to your .synopsys_dc.setup file (for synthesis) and add  
    // +incdir+$SYNOPSYS/dw/sim_ver+ to your verilog simulator command line  
    // (for simulation).  
    `include "DW01_prienc_function.inc"  
  
    input [func_A_width-1:0] func_A;  
    output [func_INDEX_width-1:0] INDEX_func;  
    assign INDEX_func = DWF_prienc(func_A);  
  
endmodule
```

HDL Usage Through Component Instantiation - VHDL

```
library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DWpackages.all;
use DWARE.DW_foundation_comp.all;

entity DW01_prienc_inst is
  generic (inst_A_width : POSITIVE := 8;
           inst_INDEX_width : POSITIVE := 4);
  port (inst_A : in std_logic_vector(inst_A_width-1 downto 0);
        INDEX_inst : out std_logic_vector(inst_INDEX_width-1 downto 0));
end DW01_prienc_inst;

architecture inst of DW01_prienc_inst is
begin

  -- Instance of DW01_prienc
  U1 : DW01_prienc
    generic map ( A_width => inst_A_width, INDEX_width => inst_INDEX_width )
    port map ( A => inst_A, INDEX => INDEX_inst );
end inst;

-- pragma translate_off
configuration DW01_prienc_inst_cfg_inst of DW01_prienc_inst is
  for inst
  end for; -- inst
end DW01_prienc_inst_cfg_inst;
-- pragma translate_on
```

HDL Usage Through Component Instantiation - Verilog

```
module DW01_prienc_inst( inst_A, INDEX_inst );

    parameter A_width = 8;
    parameter INDEX_width = 4;

    input [A_width-1 : 0] inst_A;
    output [INDEX_width-1 : 0] INDEX_inst;

    // Instance of DW01_prienc
    DW01_prienc #(A_width, INDEX_width)
        U1 ( .A(inst_A), .INDEX(INDEX_inst) );

endmodule
```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

For a version of this datasheet with visible change bars, click [here](#).

Date	Release	Updates
September 2024	DWBB_202409.0	■ Updated date and version; no technical changes
December 2023	DWBB_202312.0	■ Updated date and version; no technical changes
January 2019	DWBB_201806.5	■ Updated example in “ HDL Usage Through Component Instantiation - VHDL ” on page 5 ■ Added this Revision History table and the document links on this page

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